

SANDY BROWN

Consultants in Acoustics, Noise & Vibration

17470-R03-B

17 April 2018

Greenwich Millennium Village, Phases 3, 4 & 5 Buildings 302

Buildings 302 facade sound insulation mark ups – balconies and windows

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1 Building 302 facade sound insulation mark ups – balconies and windows

The following pages contain marked up floor plans for buildings 302 demonstrating how the design principles for opening windows and partially enclosed balconies can be applied. The minimum sound insulation performances for doors and windows on the various facades of the building are also provided.

The mark ups are based on the latest internal layouts issued by Jestico + Whiles for the building.

Table 1 provides a key to the tables for each facade mark-up in the following pages of this document.

Table 1 Key to facade mark-up table

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz} *)		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}			
XX	XX	XX	XX	XX	XX	XX	XX	XX	XX%	XX%	Yes / No	Yes / No	XX	XX	

Window / door reference number

Criterion 1: *L*_{Aeq} ≥ 45 dB - window fixed closed unless adequately shrouded

Criterion 2: *L*_{A63Hz} ≥ 40 dB - window recommended to be closed unless shrouded; predicted levels compared to Acoustic Report to determine required *D*_{63Hz} for window / door

Criteria 3 & 4: predicted levels which are compared to Acoustic Report to determine required window / door sound insulation performance

* This number is approximated from the broadband models of dredger noise, which dominates at 63 Hz. The actual level at 63 Hz is either equal to this number or as much as 3 dB lower. See discussion in cover report (17470-R02-C GMV3 Buildings 302, 303 & 304 -Internal and facade levels- acoustic Cond 87 Report, dated 17 April 2018).

All levels given are A-weighted facade noise levels in dB.

- The first level in each column (AO) is the noise level predicted with just the buildings existing ‘At Occupation’ of Buildings 302, namely Buildings 101 to 109, 201, 204 to 210 and 504.
- The second level in each column (MP) is the noise levels predicted with all the buildings of the masterplan in place – including all later parcels.

The percentages of allowable open areas are based on balcony/ shroud’s frontage only. The ends, top and floor may need to be solid, as noted herein.

Where no balcony / shroud is present, the percentage of allowable open area and the decision on whether a window may open will require further measures as noted herein. The additional screening required will be designed so that the openings are shielded from the wharf / dredger noise to ensure the internal noise levels can still be met. Please refer to the architectural drawings for more details on this.

The window / door sound insulation values given (ie, *D*_w + *C*_{tr} and *D*_{63Hz}) assume the window / door is more than 50% of the facade of the room and the remaining solid portion has at least the same performance. Where windows are a small portion of the facade and the surrounding solid part of the facade is at least 5 dB better than the windows, a slight relaxation in the requirement for the window / door as shown in Table 2 is then possible.

Table 2 Composite facade sound insulation performance, solid v glazed

Sound insulation required (from Figures on the following pages)	Minimum sound insulation of solid portion	Minimum sound insulation of fenestration if less than 20%/30%/50% of room facade
<i>D</i> _w + <i>C</i> _{tr} ≥ 30 dB	<i>D</i> _w + <i>C</i> _{tr} ≥ 35 dB	<i>D</i> _w + <i>C</i> _{tr} ≥ 27/28/29 dB
<i>D</i> _w + <i>C</i> _{tr} ≥ 25 dB	<i>D</i> _w + <i>C</i> _{tr} ≥ 30 dB	<i>D</i> _w + <i>C</i> _{tr} ≥ 22/23/24 dB
<i>D</i> _w + <i>C</i> _{tr} ≥ 20 dB	<i>D</i> _w + <i>C</i> _{tr} ≥ 25 dB	<i>D</i> _w + <i>C</i> _{tr} ≥ 17/18/19 dB

2 Building 302 Block A – Ground level

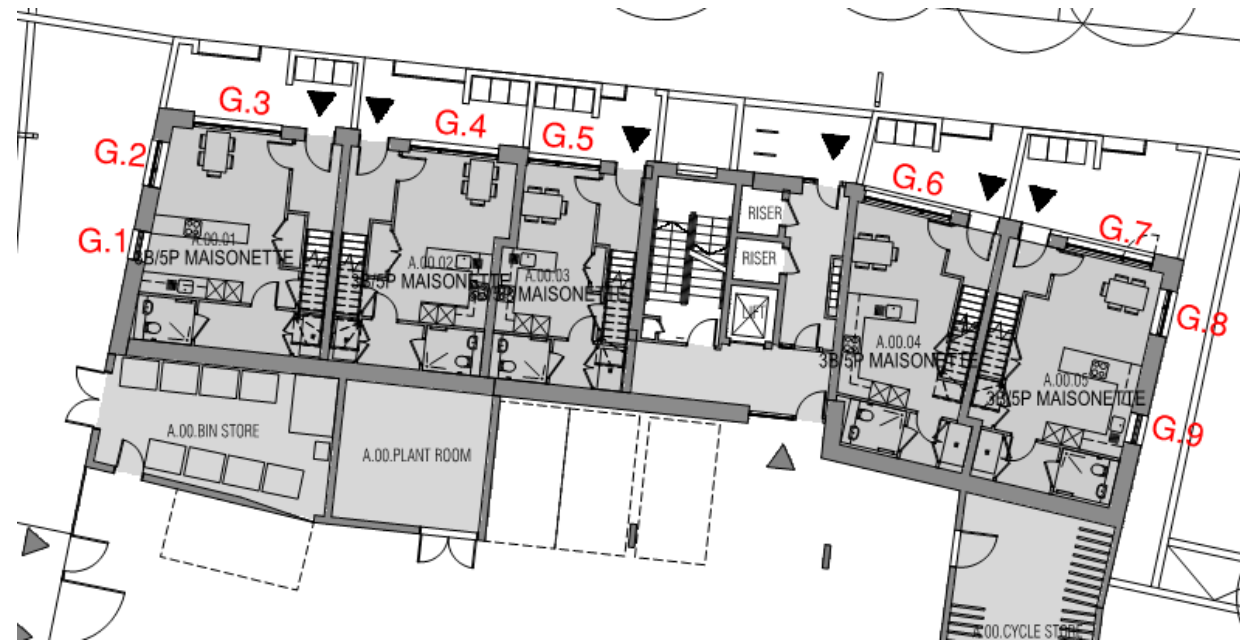


Figure 1 Building 302 Block A – Ground level floor plan

Table 3 Building 302 Block A – Ground level predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (L_{Aeq})		Dredgers (L_{A63Hz})		All sources night (L_{Aeq})		All sources day (L_{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	D_w+C_{tr}	D_{63Hz}	
G.1	35	37	33	28	57	56	60	59	100%	100%	YES	YES	28	10	
G.2	35	37	33	28	57	56	61	60	100%	100%	YES	YES	28	10	
G.3	41	40	37	32	61	61	62	62	100%	100%	YES	YES	32	15	
G.4	41	40	36	33	61	61	62	62	100%	100%	YES	YES	32	15	
G.5	40	40	35	35	61	61	62	62	100%	100%	YES	YES	32	15	
G.6	40	40	34	34	61	61	62	62	100%	100%	YES	YES	32	15	
G.7	40	40	34	34	61	61	62	62	100%	100%	YES	YES	27	15	
G.8	40	40	36	34	55	55	56	54	100%	100%	YES	YES	26	15	
G.9	40	40	36	34	55	55	56	54	100%	100%	YES	YES	26	15	

3 Building 302 Block B – Ground level

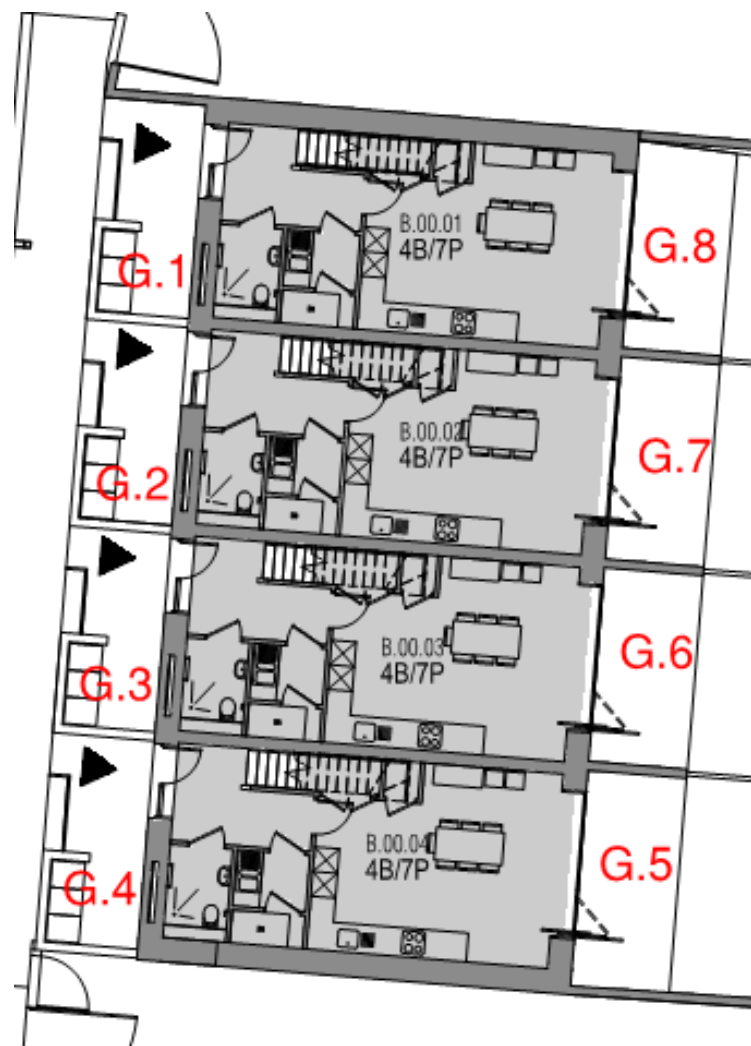


Figure 2 Building 302 Block B – Ground level floor plan

Table 4 Building 302 Block B – Ground level predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{Tr}	<i>D</i> _{63Hz}	
G.1	35	39	31	32	52	45	59	52	100%	100%	YES	YES	18	15	
G.2	35	39	31	32	52	45	59	51	100%	100%	YES	YES	18	15	
G.3	35	38	31	33	51	44	59	51	100%	100%	YES	YES	17	15	
G.4	35	38	31	34	51	44	58	50	100%	100%	YES	YES	17	15	
G.5	41	42	35	36	43	43	50	47	100%	100%	YES	YES	14	15	
G.6	41	41	35	35	44	43	51	48	100%	100%	YES	YES	15	15	
G.7	41	41	35	35	43	42	50	48	100%	100%	YES	YES	14	15	
G.8	40	41	35	35	44	42	50	47	100%	100%	YES	YES	15	15	

4 Building 302 Block C – Ground level

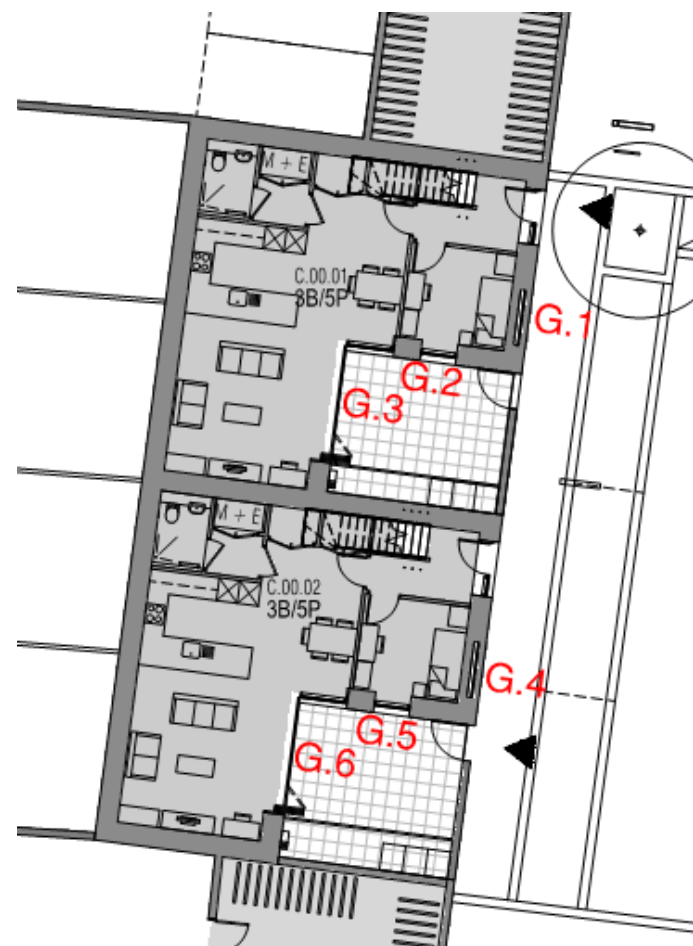


Figure 3 Building 302 Block C – Ground level floor plan

Table 5 Building 302 Block C – Ground level predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (L_{Aeq})		Dredgers (L_{A63Hz})		All sources night (L_{Aeq})		All sources day (L_{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	D_w+C_{tr}	D_{63Hz}	
G.1	40	41	34	34	47	45	53	0	100%	100%	YES	YES	18	15	
G.2	40	41	32	32	46	43	52	47	100%	100%	YES	YES	17	15	
G.3	40	41	32	32	46	43	52	47	100%	100%	YES	YES	17	15	
G.4	41	42	33	33	47	45	54	49	100%	100%	YES	YES	18	15	
G.5	42	42	31	32	49	44	56	48	100%	100%	YES	YES	20	15	
G.6	42	42	31	32	49	44	56	48	100%	100%	YES	YES	20	15	

5 Building 302 Block D – Ground level

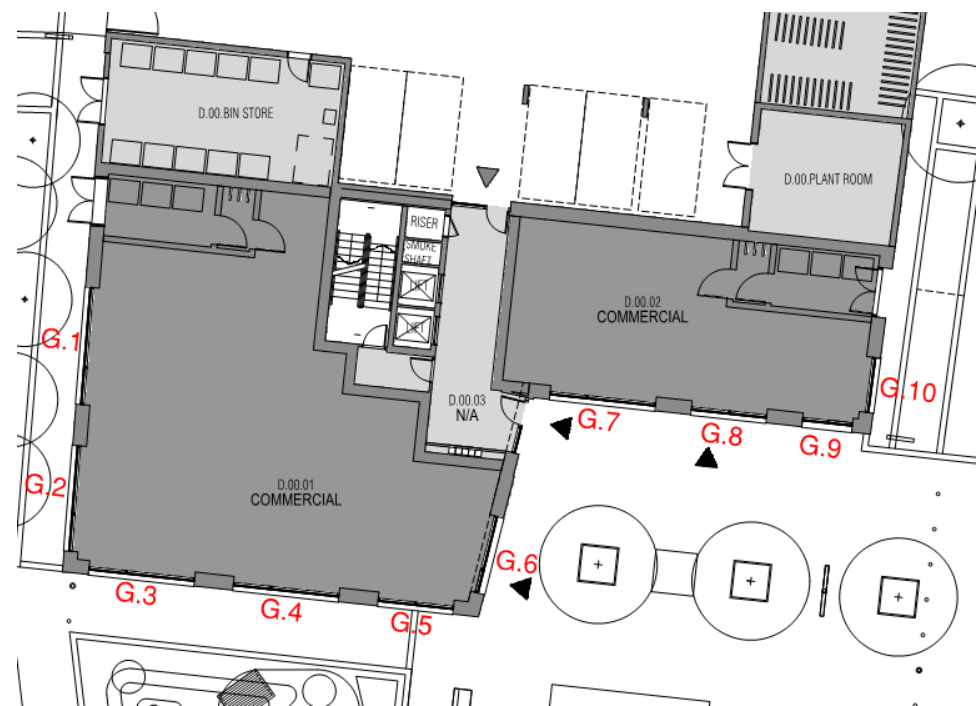


Figure 4 Building 302 Block D – Ground level floor plan

Table 6 Building 302 Block D – Ground level predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}			
G.1	34	40	29	31	52	44	59	50	n/a	n/a	YES	YES	23	8	
G.2	34	40	29	31	52	44	59	50	n/a	n/a	YES	YES	23	8	
G.3	46	43	29	33	55	48	62	55	n/a	n/a	YES	YES	26	15	
G.4	46	43	28	33	55	48	62	55	n/a	n/a	YES	YES	26	15	
G.5	46	43	28	35	55	48	62	55	n/a	n/a	YES	YES	26	15	
G.6	46	43	28	35	55	48	62	55	n/a	n/a	YES	YES	26	15	
G.7	46	41	28	35	55	47	62	55	n/a	n/a	YES	YES	26	15	
G.8	46	41	28	35	55	47	62	55	n/a	n/a	YES	YES	26	15	
G.9	46	41	28	35	55	46	62	55	n/a	n/a	YES	YES	26	15	
G.10	44	42	34	36	53	47	64	58	n/a	n/a	YES	YES	25	13	

6 Building 302 Block A – Level 1

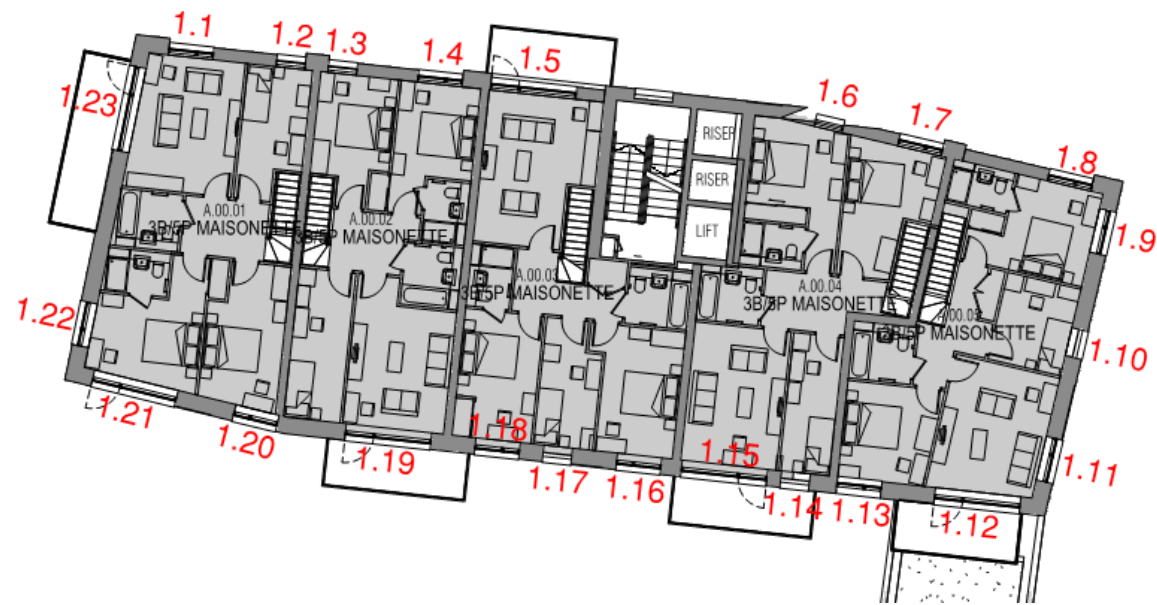


Figure 5 Building 302 Block A – Level 1 floor plan

Table 7 Building 302 Block A – Level 1 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}			
1.1	42	41	38	32	60	60	61	61	100%	100%	YES	YES	31	15	
1.2	42	41	37	32	61	61	62	62	100%	100%	YES	YES	32	15	
1.3	42	41	37	33	60	61	62	62	100%	100%	YES	YES	32	15	
1.4	41	41	36	31	61	61	62	62	100%	100%	YES	YES	32	15	
1.5	41	41	36	34	61	61	62	62	100%	100%	YES	YES	32	15	
1.6	42	41	36	35	60	60	61	62	100%	100%	YES	YES	31	15	
1.7	42	41	35	34	60	60	61	61	100%	100%	YES	YES	31	15	
1.8	40	41	35	35	60	60	61	61	100%	100%	YES	YES	31	15	
1.9	41	41	36	36	55	55	56	55	100%	100%	YES	YES	21	15	
1.10	41	41	35	35	54	54	55	54	100%	100%	YES	YES	25	15	
1.11	40	42	35	35	48	46	54	50	100%	100%	YES	YES	19	15	
1.12	39	41	35	33	47	45	54	49	100%	100%	YES	YES	20	15	
1.13	39	41	32	34	47	45	54	49	100%	100%	YES	YES	20	15	
1.14	39	41	33	36	48	45	55	49	100%	100%	YES	YES	19	15	
1.15	39	41	30	34	49	44	56	50	100%	100%	YES	YES	22	15	
1.16	39	41	32	33	49	45	57	50	100%	100%	YES	YES	23	15	
1.17	39	41	34	33	50	46	57	52	100%	100%	YES	YES	23	15	
1.18	39	41	34	34	49	46	57	51	100%	100%	YES	YES	20	15	
1.19	39	41	32	34	49	45	57	50	100%	100%	YES	YES	23	15	
1.20	39	41	32	34	49	45	56	50	100%	100%	YES	YES	20	15	
1.21	39	41	33	33	51	48	56	54	100%	100%	YES	YES	22	15	
1.22	34	39	32	30	56	55	60	59	100%	100%	YES	YES	27	15	
1.23	37	38	36	30	58	58	61	60	100%	100%	YES	YES	29	15	

7 Building 302 Block B – Level 1



Figure 6 Building 302 Block B – Level 1 floor plan

Table 8 Building 302 Block B – Level 1 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _W + <i>C</i> _{Tr}	<i>D</i> _{63Hz}	
1.1	35	40	31	33	52	47	59	53	100%	100%	YES	YES	23	15	
1.2	35	40	31	33	52	46	59	53	100%	100%	YES	YES	23	15	
1.3	35	40	31	34	52	45	59	51	100%	100%	YES	YES	23	15	
1.4	36	39	32	35	52	45	59	51	100%	100%	YES	YES	23	15	
1.5	44	44	36	37	45	45	50	49	100%	100%	YES	YES	16	15	
1.6	44	44	36	37	46	45	52	49	100%	100%	YES	YES	18	15	
1.7	44	44	36	36	47	45	52	49	100%	100%	YES	YES	18	15	
1.8	43	44	36	36	46	45	52	49	100%	100%	YES	YES	18	15	
1.9	43	44	36	36	46	45	52	49	100%	100%	YES	YES	18	15	
1.10	43	44	35	36	46	45	52	49	100%	100%	YES	YES	18	15	
1.11	42	43	35	35	46	44	51	47	100%	100%	YES	YES	17	15	
1.12	42	43	36	35	46	44	52	48	100%	100%	YES	YES	18	15	

8 Building 302 Block C – Level 1



Figure 7 Building 302 Block C – Level 1 floor plan

Table 9 Building 302 Block C – Level 1 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}			
1.1	41	42	35	35	49	46	54	51	100%	100%	YES	YES	20	15	
1.2	41	42	35	35	48	46	54	51	100%	100%	YES	YES	20	15	
1.3	41	42	34	34	48	45	54	49	100%	100%	YES	YES	19	15	
1.4	41	42	34	34	48	45	54	49	100%	100%	YES	YES	20	15	
1.5	41	42	34	34	48	45	55	50	100%	100%	YES	YES	21	15	
1.6	41	42	34	34	48	45	55	50	100%	100%	YES	YES	21	15	
1.7	41	43	33	34	50	45	56	48	100%	100%	YES	YES	16	15	
1.8	41	43	35	36	50	44	56	48	100%	100%	YES	YES	22	15	

9 Building 302 Block D – Level 1



Figure 8 Building 302 Block D – Level 1 floor plan

Table 10 Building 302 Block D – Level 1 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}			
1.1	42	43	35	37	50	45	57	50	100%	100%	YES	YES	21	15	
1.2	42	43	35	37	50	45	57	48	100%	100%	YES	YES	21	15	
1.3	43	43	36	36	47	45	56	48	100%	100%	YES	YES	22	15	
1.4	42	43	35	36	48	45	54	48	100%	100%	YES	YES	20	15	
1.5	43	43	37	37	46	45	54	49	100%	100%	YES	YES	20	15	

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}	
1.6	42	43	37	38	46	45	52	49	100%	100%	YES	YES	18	15	Seal and ventilate via balcony or provide Juliet screening arrangement
1.7	42	43	35	36	46	45	52	49	100%	100%	YES	YES	18	15	
1.8	42	43	35	36	46	45	52	49	100%	100%	YES	YES	17	15	
1.9	41	42	35	35	46	44	53	49	100%	100%	YES	YES	17	15	
1.10	42	42	35	35	52	45	61	50	100%	100%	YES	YES	23	15	
1.11	40	42	36	37	52	45	61	51	100%	100%	YES	YES	23	15	
1.12	42	43	35	38	53	47	64	57	100%	100%	YES	YES	25	15	
1.13	45	43	35	36	53	47	64	58	85%	100%	YES	YES	25	15	
1.14	45	43	29	33	55	46	62	56	85%	100%	YES	YES	26	15	
1.15	46	42	29	35	56	47	62	56	70%	100%	YES	YES	27	15	
1.16	46	42	29	36	56	47	62	56	70%	100%	YES	YES	28	15	
1.17	46	42	29	36	56	47	62	56	70%	100%	YES	YES	28	15	
1.18	46	43	29	37	55	47	62	55	70%	100%	YES	YES	28	15	
1.19	46	43	30	36	55	47	63	55	70%	100%	YES	YES	26	15	
1.20	45	44	29	35	55	48	62	55	85%	100%	YES	YES	26	15	
1.21	46	44	29	35	55	49	62	55	70%	100%	YES	YES	28	15	
1.22	46	44	29	33	55	49	62	55	70%	100%	YES	YES	28	15	
1.23	46	43	29	31	55	49	62	55	70%	100%	YES	YES	26	15	
1.24	35	41	28	32	53	46	60	52	100%	100%	YES	YES	24	15	
1.25	34	40	28	32	52	44	59	51	100%	100%	YES	YES	23	15	
1.26	34	40	28	32	52	44	59	50	100%	100%	YES	YES	25	15	
1.27	34	41	29	32	52	44	59	50	100%	100%	YES	YES	25	15	
1.28	33	40	29	32	52	44	59	50	100%	100%	YES	YES	23	15	
1.29	35	39	29	35	52	44	59	50	100%	100%	YES	YES	23	15	

10 Building 302 Block A – Level 2

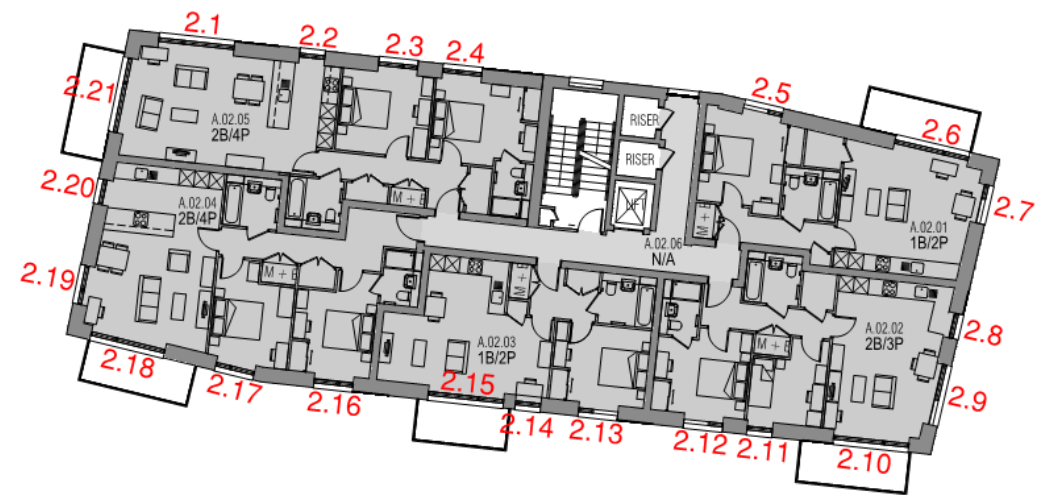


Figure 9 Building 302 Block A – Level 2 floor plan

Table 11 Building 302 Block A – Level 2 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}	
2.1	43	41	39	31	60	60	61	61	100%	100%	YES	YES	31	16	
2.2	43	41	37	33	60	60	61	61	100%	100%	YES	YES	31	15	
2.3	42	42	37	34	60	60	61	62	100%	100%	YES	YES	31	15	
2.4	42	42	37	34	60	60	61	62	100%	100%	YES	YES	31	15	
2.5	42	42	36	35	60	60	61	61	100%	100%	YES	YES	31	15	
2.6	41	41	36	35	60	60	61	61	100%	100%	YES	YES	31	15	
2.7	40	41	35	35	57	56	58	57	100%	100%	YES	YES	28	15	
2.8	40	41	37	35	56	56	58	56	100%	100%	YES	YES	27	15	
2.9	41	42	37	37	55	54	57	55	100%	100%	YES	YES	26	15	
2.10	42	42	38	36	54	54	56	55	100%	100%	YES	YES	25	15	
2.11	41	43	37	37	49	46	55	50	100%	100%	YES	YES	20	15	
2.12	41	43	36	37	49	46	55	50	100%	100%	YES	YES	21	15	
2.13	41	42	36	36	49	45	55	50	100%	100%	YES	YES	21	15	
2.14	39	42	30	36	49	45	56	50	100%	100%	YES	YES	22	15	
2.15	39	42	33	34	50	45	57	52	100%	100%	YES	YES	21	15	
2.16	39	42	34	34	50	45	57	51	100%	100%	YES	YES	23	15	
2.17	39	41	34	34	49	45	56	51	100%	100%	YES	YES	22	15	
2.18	39	41	33	33	51	46	56	51	100%	100%	YES	YES	22	15	
2.19	35	40	32	31	56	56	60	59	100%	100%	YES	YES	27	15	
2.20	35	39	32	29	57	57	60	59	100%	100%	YES	YES	28	15	
2.21	39	38	38	31	58	57	61	60	100%	100%	YES	YES	29	15	

Building 302 Block B – Level 2

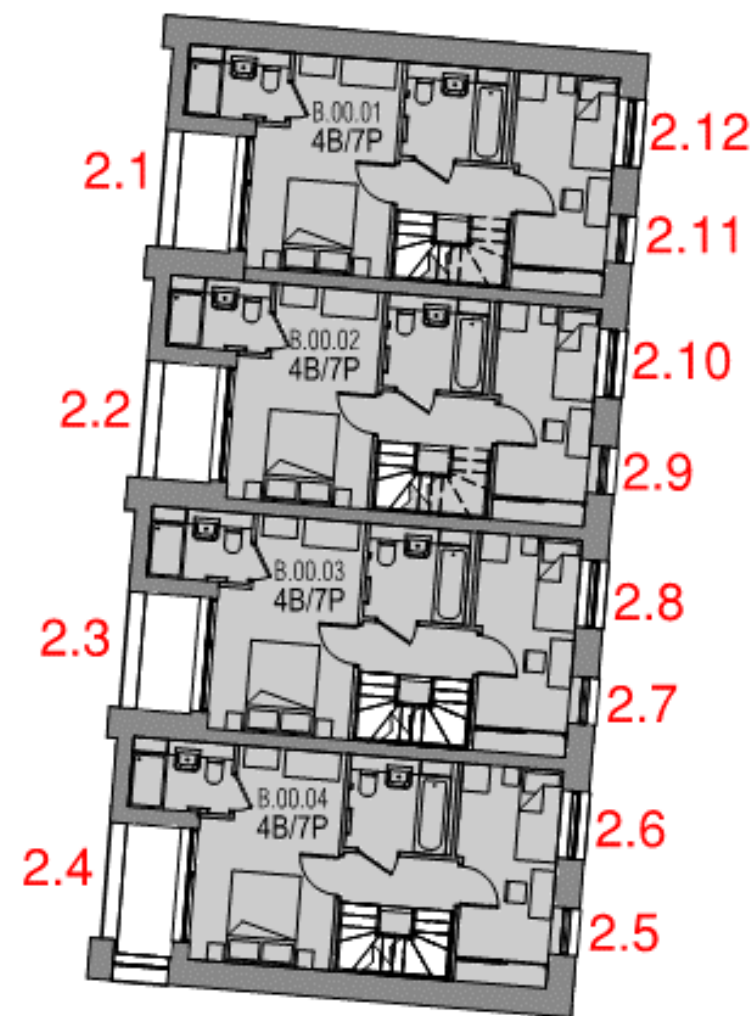


Figure 10 Building 302 Block B – Level 2 floor plan

Table 12 Building 302 Block B – Level 2 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}			
2.1	37	42	33	34	53	47	60	54	100%	100%	YES	YES	26	15	Juliet screening arrangement
2.2	37	43	33	35	53	47	60	52	100%	100%	YES	YES	26	15	
2.3	37	43	33	35	52	47	60	52	100%	100%	YES	YES	26	15	
2.4	37	42	33	36	52	46	59	52	100%	100%	YES	YES	25	15	
2.5	44	44	37	37	47	45	52	50	100%	100%	YES	YES	18	15	
2.6	44	45	37	37	47	46	53	50	100%	85%	YES	YES	18	15	
2.7	44	45	37	37	47	46	53	50	100%	85%	YES	YES	18	15	
2.8	44	44	36	37	47	45	53	50	100%	85%	YES	YES	18	15	
2.9	44	44	36	37	47	45	53	50	100%	100%	YES	NO	18	15	
2.10	44	44	36	36	47	45	53	50	100%	100%	YES	YES	18	15	
2.11	43	43	36	36	47	45	52	49	100%	100%	YES	YES	18	15	
2.12	43	43	36	36	47	45	53	48	100%	100%	YES	YES	18	15	

11 Building 302 Block D – Level 2

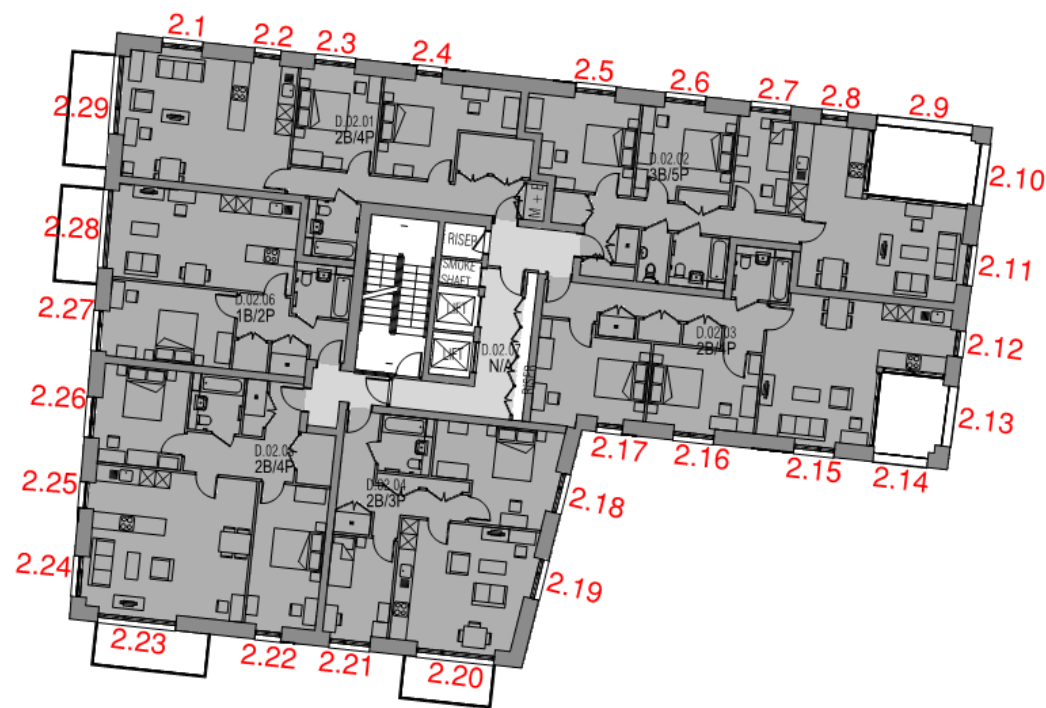


Figure 11 Building 302 Block D – Level 2 floor plan

Table 13 Building 302 Block D – Level 2 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}			
2.1	43	44	37	38	51	46	58	51	100%	100%	YES	YES	22	15	
2.2	43	44	37	38	50	45	57	50	100%	100%	YES	YES	21	15	
2.3	43	44	37	38	48	45	54	49	100%	100%	YES	YES	20	15	
2.4	43	44	36	37	49	45	55	49	100%	100%	YES	YES	21	15	
2.5	43	44	36	37	49	45	55	50	100%	100%	YES	YES	21	15	
2.6	43	44	37	37	47	45	54	50	100%	100%	YES	YES	20	15	

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}	
2.7	43	44	36	37	47	45	53	50	100%	100%	YES	YES	19	15	Seal and ventilate via balcony or provide Juliet screening arrangement
2.8	43	44	36	37	47	45	53	50	100%	100%	YES	YES	18	15	
2.9	42	44	36	37	46	45	53	50	100%	100%	YES	YES	17	15	
2.10	41	42	35	35	51	45	60	50	100%	100%	YES	YES	22	15	
2.11	41	42	36	37	52	45	60	52	100%	100%	YES	YES	23	15	
2.12	42	43	36	37	53	45	63	51	100%	100%	YES	YES	24	15	
2.13	45	44	36	36	53	48	63	59	85%	100%	YES	YES	24	15	
2.14	45	44	29	33	55	47	62	56	85%	100%	YES	YES	26	15	
2.15	46	42	29	36	56	47	63	56	70%	100%	YES	YES	27	15	
2.16	46	42	28	36	56	48	63	56	70%	100%	YES	YES	29	15	
2.17	46	43	28	37	56	48	62	56	70%	100%	YES	YES	28	15	
2.18	46	43	29	37	56	48	62	56	70%	100%	YES	YES	28	15	
2.19	47	44	30	37	56	48	63	55	45%	100%	YES	YES	27	15	
2.20	46	45	28	36	55	49	62	56	70%	100%	YES	YES	26	15	
2.21	46	45	28	36	55	49	62	56	70%	100%	YES	YES	28	15	
2.22	46	44	28	34	55	49	62	56	70%	100%	YES	YES	28	15	
2.23	46	43	28	34	55	49	62	55	70%	100%	YES	YES	26	15	
2.24	35	42	28	32	53	47	60	52	100%	100%	YES	YES	24	15	
2.25	34	41	29	32	53	45	60	51	100%	100%	YES	YES	24	15	
2.26	34	41	29	32	53	45	60	51	100%	100%	YES	YES	26	15	
2.27	34	41	29	32	53	44	60	50	100%	100%	YES	YES	26	15	
2.28	34	41	29	32	53	44	60	50	100%	100%	YES	YES	24	15	
2.29	35	40	32	36	53	45	60	51	100%	100%	YES	YES	24	15	

12 Building 302 Block A – Level 3

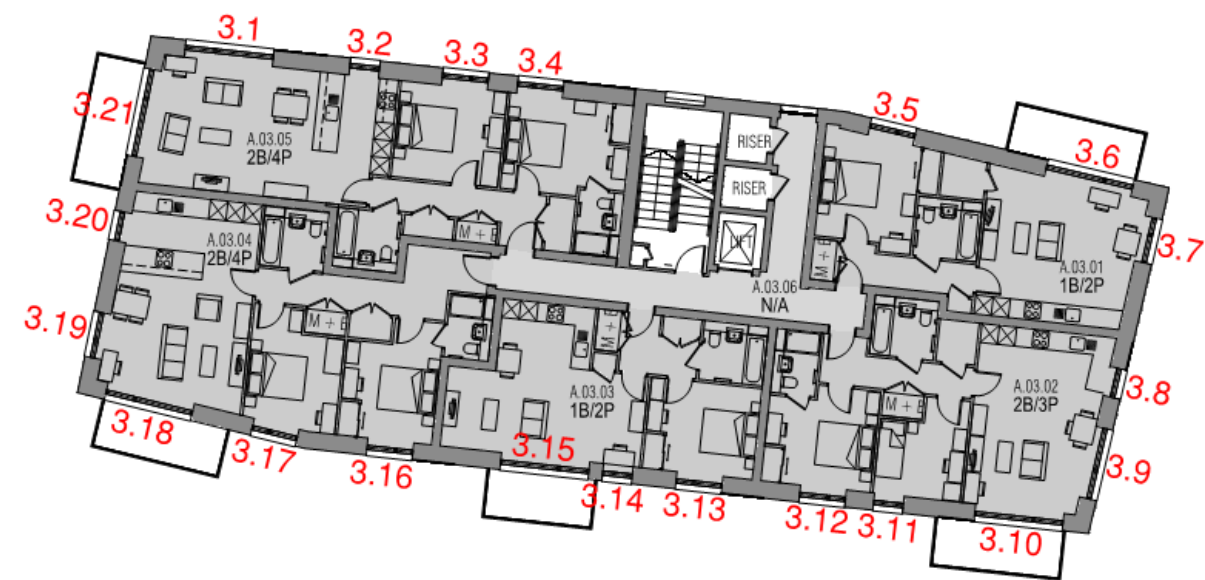


Figure 12 Building 302 Block A – Level 3 floor plan

Table 14 Building 302 Block A – Level 3 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}	
3.1	45	42	42	34	59	59	61	61	85%	100%	YES	YES	30	19	Seal and ventilate via balcony or provide Juliet screening arrangement
3.2	44	42	40	34	59	60	61	61	85%	100%	YES	YES	31	17	Juliet screening arrangement
3.3	43	42	40	36	59	60	61	61	85%	100%	YES	YES	31	17	Juliet screening arrangement
3.4	43	42	40	36	59	60	61	61	85%	100%	YES	YES	31	17	Juliet screening arrangement
3.5	42	42	39	36	59	59	60	61	100%	100%	YES	YES	30	16	
3.6	41	42	36	36	59	59	60	61	100%	100%	YES	YES	30	15	
3.7	41	41	36	36	56	56	58	57	100%	100%	YES	YES	27	15	
3.8	41	42	36	36	56	55	58	56	100%	100%	YES	YES	27	15	
3.9	43	43	37	38	54	54	57	55	100%	100%	YES	YES	25	15	
3.10	42	43	38	38	49	46	56	51	100%	100%	YES	YES	20	15	
3.11	41	43	36	38	49	46	56	50	100%	100%	YES	YES	22	15	
3.12	41	43	36	36	50	45	56	50	100%	100%	YES	YES	22	15	
3.13	40	42	30	36	50	46	57	50	100%	100%	YES	YES	23	15	
3.14	40	42	30	36	50	46	57	50	100%	100%	YES	YES	23	15	
3.15	40	42	34	34	50	46	57	51	100%	100%	YES	YES	21	15	
3.16	40	42	35	34	50	46	57	52	100%	100%	YES	YES	23	15	
3.17	41	43	38	38	50	46	57	52	100%	100%	YES	YES	23	15	
3.18	41	43	39	38	51	46	57	52	100%	100%	YES	YES	22	16	
3.19	36	41	33	33	56	55	60	59	100%	100%	YES	YES	27	15	
3.20	35	40	32	29	56	56	60	59	100%	100%	YES	YES	27	15	
3.21	41	39	32	31	57	57	61	59	100%	100%	YES	YES	28	15	

13 Building 302 Block B – Level 3



Figure 13 Building 302 Block B – Level 3 floor plan

Table 15 Building 302 Block B – Level 3 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}			
3.1	42	44	37	38	53	48	60	54	100%	100%	YES	YES	26	15	Juliet screening arrangement
3.2	42	45	36	37	53	48	60	53	100%	85%	YES	YES	26	15	
3.3	42	45	36	37	53	48	60	53	100%	85%	YES	YES	26	15	
3.4	42	45	36	38	53	47	60	53	100%	85%	YES	YES	26	15	
3.5	43	44	36	37	48	45	54	50	100%	100%	YES	YES	14	15	Juliet screening arrangement
3.6	43	44	36	37	48	45	54	50	100%	100%	YES	YES	14	15	
3.7	43	43	36	37	48	45	54	50	100%	100%	YES	YES	14	15	
3.8	42	43	36	38	48	45	54	50	100%	100%	YES	YES	14	15	

14 Building 302 Block D – Level 3

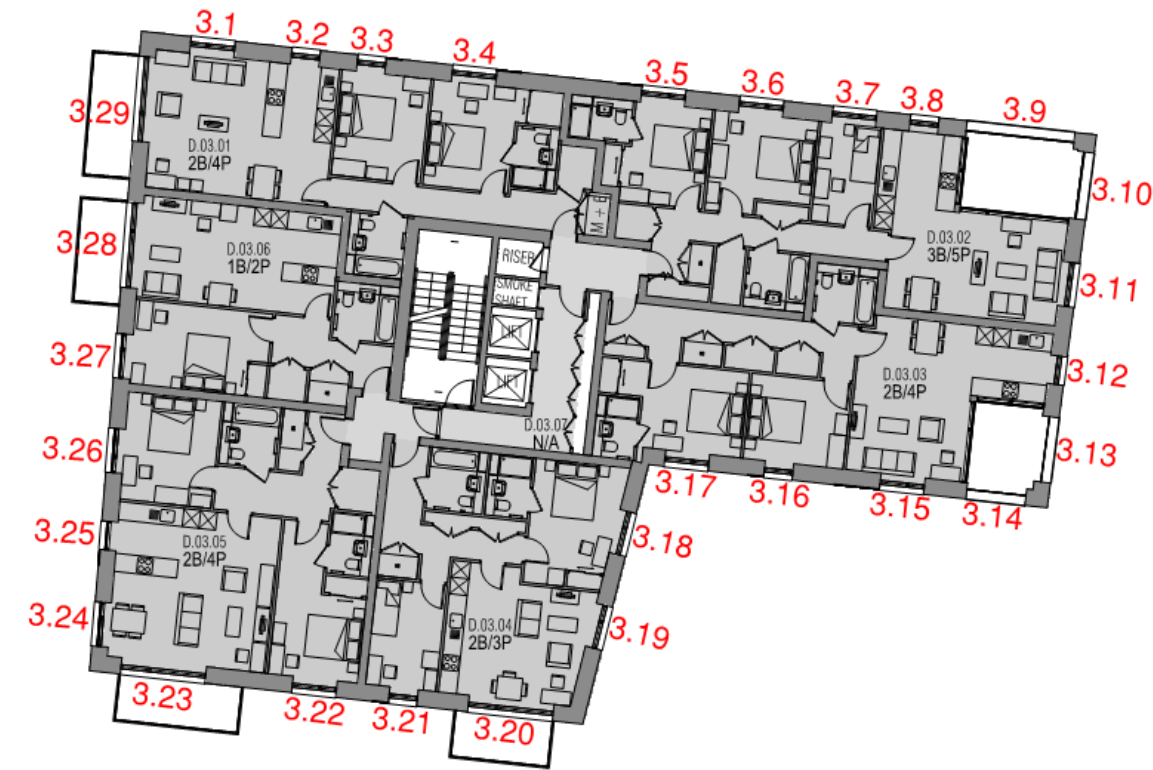


Figure 14 Building 302 Block D – Level 3 floor plan

Table 16 Building 302 Block D – Level 3 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (L_{Aeq})		Dredgers (L_{A63Hz})		All sources night (L_{Aeq})		All sources day (L_{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	D_w+C_{tr}	D_{63Hz}			
3.1	44	45	38	39	51	47	58	51	100%	85%	YES	YES	22	16	Ventilate via balcony or provide Juliet screening arrangement
3.2	44	45	37	39	51	46	58	51	100%	85%	YES	YES	22	16	Juliet screening arrangement
3.3	44	44	37	38	50	46	57	50	100%	100%	YES	YES	23	15	
3.4	44	44	37	38	50	46	57	50	100%	100%	YES	YES	23	15	
3.5	44	44	37	38	49	46	55	50	100%	100%	YES	YES	21	15	
3.6	44	44	37	38	49	46	55	50	100%	100%	YES	YES	21	15	

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}	
3.7	44	44	37	38	48	46	55	50	100%	100%	YES	YES	21	15	Seal and ventilate via balcony or provide Juliet screening arrangement 70% open area with 1 dB low frequency allowance
3.8	44	44	37	38	48	46	55	50	100%	100%	YES	YES	19	15	
3.9	44	43	37	37	48	46	55	51	100%	100%	YES	YES	19	15	
3.10	43	44	37	38	48	46	61	52	100%	100%	YES	YES	22	15	
3.11	43	44	37	38	52	46	61	53	100%	100%	YES	YES	23	15	
3.12	45	46	40	42	53	48	65	58	85%	45%	YES	YES	26	19	Seal and ventilate via balcony or provide Juliet screening arrangement
3.13	46	45	40	41	54	48	65	59	70%	70%	YES	YES	26	18	
3.14	45	43	29	34	56	47	63	56	85%	100%	YES	YES	27	15	
3.15	46	42	28	34	56	47	63	58	70%	100%	YES	YES	27	15	
3.16	46	43	28	39	56	48	63	56	70%	100%	YES	YES	29	16	
3.17	46	44	28	40	56	48	63	56	70%	85%	YES	NO	29	17	Juliet screening arrangement
3.18	47	44	31	40	56	48	63	56	45%	85%	YES	NO	29	17	Juliet screening arrangement
3.19	47	44	32	38	56	48	63	56	45%	100%	YES	YES	27	15	Seal and ventilate via balcony or provide Juliet screening arrangement
3.20	45	45	28	39	56	49	63	56	85%	85%	YES	YES	27	16	Juliet screening arrangement
3.21	46	45	28	39	56	49	63	56	70%	85%	YES	YES	29	16	
3.22	46	44	28	37	56	49	63	56	70%	100%	YES	YES	29	15	
3.23	46	44	28	35	56	49	62	56	70%	100%	YES	YES	27	15	
3.24	35	44	28	37	53	47	60	53	100%	100%	YES	YES	24	15	
3.25	35	43	28	37	53	46	60	52	100%	100%	YES	YES	24	15	
3.26	34	42	28	33	53	46	60	52	100%	100%	YES	YES	26	15	
3.27	34	41	28	32	53	44	60	50	100%	100%	YES	YES	26	15	
3.28	34	41	28	33	53	45	60	50	100%	100%	YES	YES	24	15	
3.29	34	41	28	33	53	45	60	51	100%	100%	YES	YES	24	15	

15 Building 302 Block A – Level 4

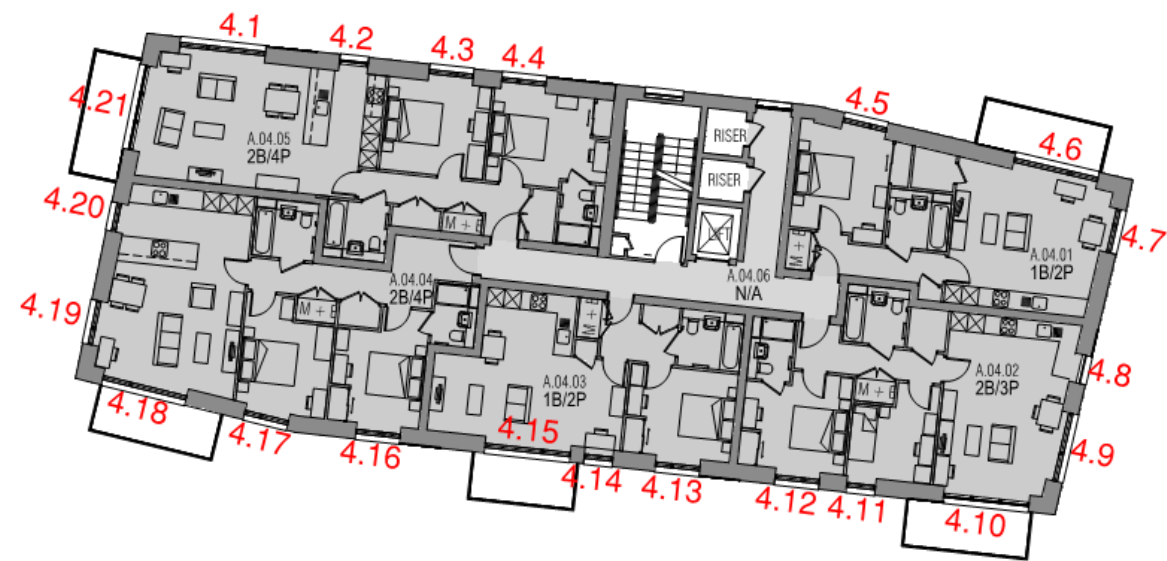


Figure 15 Building 302 Block A – Level 4 floor plan

Table 17 Building 302 Block A – Level 4 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}			
4.1	45	43	42	35	59	59	61	61	85%	100%	YES	YES	30	19	Seal and ventilate via balcony or provide Juliet screening arrangement
4.2	45	43	42	35	59	59	61	61	45%	100%	YES	YES	30	19	Juliet screening arrangement. 70% open area with 1 dB low frequency allowance
4.3	44	43	41	36	58	59	61	61	70%	100%	YES	YES	30	18	Juliet screening arrangement
4.4	44	43	41	37	58	59	60	61	70%	100%	YES	YES	30	18	Juliet screening arrangement
4.5	43	43	39	37	58	58	60	60	100%	100%	YES	YES	29	16	
4.6	43	43	37	37	58	58	60	60	100%	100%	YES	YES	29	15	
4.7	42	42	37	37	55	55	58	56	100%	100%	YES	YES	26	15	
4.8	44	44	39	39	53	54	55	55	100%	100%	YES	YES	25	16	
4.9	44	44	39	39	53	54	55	55	100%	100%	YES	YES	25	16	
4.10	44	45	41	41	50	47	56	52	70%	70%	YES	YES	21	18	
4.11	42	43	37	37	50	46	56	51	100%	100%	YES	YES	22	15	
4.12	42	43	37	37	50	46	57	51	100%	100%	YES	YES	23	15	
4.13	40	42	37	37	50	45	57	51	100%	100%	YES	YES	23	15	
4.14	40	42	37	37	50	45	57	51	100%	100%	YES	YES	23	15	
4.15	41	44	37	39	50	45	57	52	100%	100%	YES	YES	21	16	
4.16	42	44	39	39	50	46	57	52	100%	100%	YES	YES	23	16	
4.17	42	44	39	39	50	45	57	52	100%	100%	YES	YES	23	16	
4.18	42	43	39	39	51	47	57	53	100%	100%	YES	YES	22	16	
4.19	38	42	37	37	55	54	60	59	100%	100%	YES	YES	26	15	
4.20	35	40	31	30	56	55	60	59	100%	100%	YES	YES	27	15	
4.21	41	40	31	29	56	56	60	59	100%	100%	YES	YES	27	15	

16 Building 302 Block D – Level 4

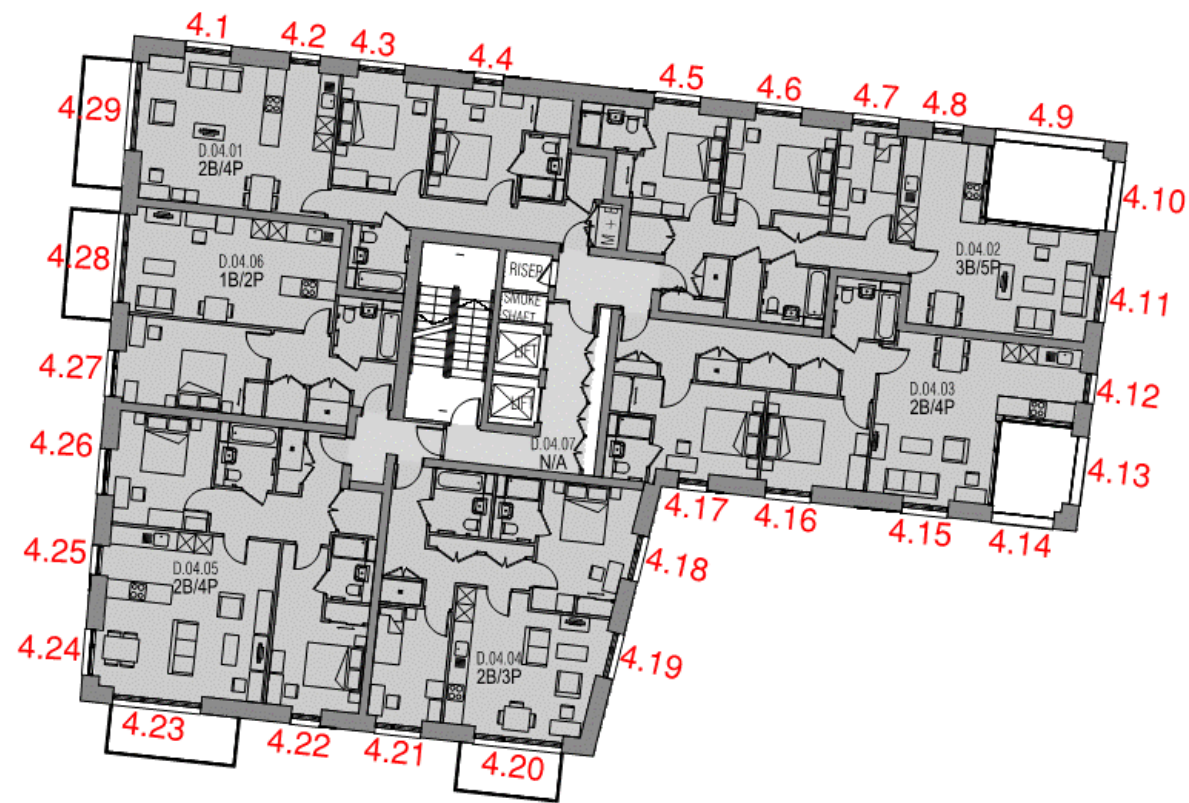


Figure 16 Building 302 Block D – Level 4 floor plan

Table 18 Building 302 Block D – Level 4 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (L_{Aeq})		Dredgers (L_{A63Hz})		All sources night (L_{Aeq})		All sources day (L_{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	D_w+C_{tr}	D_{63Hz}			
4.1	45	47	40	43	51	48	58	52	85%	30%	YES	YES	22	20	Seal and ventilate via balcony or provide Juliet screening arrangement. 45% open area with 1 dB low frequency allowance
4.2	45	47	40	43	51	48	58	52	85%	30%	YES	YES	22	20	Juliet screening arrangement. 45% open area with 1 dB low frequency allowance
4.3	45	46	40	43	51	48	58	52	85%	30%	YES	YES	24	20	Juliet screening arrangement 70% open area with 2 dB low frequency allowance
4.4	45	46	40	42	51	48	58	51	85%	45%	YES	YES	24	19	Juliet screening arrangement 70% open area with 1 dB low frequency allowance
4.5	46	47	42	43	51	48	58	51	45%	30%	YES	YES	24	20	Juliet screening arrangement 45% open area with 1 dB low frequency allowance
4.6	46	47	42	43	51	48	57	52	45%	30%	YES	YES	23	20	Juliet screening arrangement 45% open area with 1 dB low frequency allowance
4.7	46	47	42	42	51	48	57	52	45%	45%	YES	YES	23	19	Juliet screening arrangement
4.8	45	47	42	42	51	47	57	52	45%	45%	YES	YES	22	19	Seal and ventilate via balcony or provide Juliet screening arrangement.
4.9	44	46	40	40	51	47	58	53	85%	70%	YES	YES	22	17	
4.10	46	46	41	41	53	48	61	54	70%	70%	YES	YES	24	18	
4.11	45	46	42	43	54	48	61	54	45%	30%	YES	Yes	25	20	Seal and ventilate via balcony or provide Juliet screening arrangement. 70% open area with 2 dB low frequency allowance
4.12	46	47	41	43	54	49	65	58	70%	30%	YES	YES	26	21	Seal and ventilate via balcony or provide Juliet screening arrangement. 70% open area with 2 dB low frequency allowance
4.13	47	47	41	43	56	49	63	58	45%	30%	YES	YES	27	20	45% open area with 1 dB low frequency allowance
4.14	46	45	30	39	56	48	63	57	70%	85%	YES	YES	27	16	
4.15	46	44	30	39	56	48	63	57	70%	100%	YES	YES	27	16	Seal and ventilate via balcony or provide Juliet screening arrangement.

Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes	
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})									
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}				
4.16	46	45	30	42	56	49	63	57	70%	85%	YES	YES	29	19	Juliet screening arrangement	
4.17	46	45	30	42	56	49	63	57	70%	85%	YES	YES	29	19	Juliet screening arrangement	
4.18	47	46	33	43	56	49	63	56	45%	70%	YES	YES	29	20	Juliet screening arrangement	
4.19	47	46	34	43	56	49	63	56	45%	30%	YES	YES	27	20	Seal and ventilate via balcony or provide Juliet screening arrangement. 45% open area with 1 dB low frequency allowance	
4.20	45	47	30	42	56	50	63	57	85%	45%	YES	YES	27	19		
4.21	46	47	30	42	56	50	63	57	70%	100%	YES	YES	29	19		Juliet screening arrangement
4.22	46	45	30	40	56	50	63	57	100%	100%	YES	YES	29	17		Juliet screening arrangement
4.23	46	44	30	39	56	50	63	56	70%	100%	YES	YES	27	16		
4.24	36	44	30	37	53	48	60	53	100%	100%	YES	YES	24	15		
4.25	36	44	30	37	53	47	60	52	100%	100%	YES	YES	24	15		
4.26	36	44	30	37	53	47	60	52	100%	100%	YES	YES	26	15		
4.27	35	41	30	33	53	45	60	52	100%	100%	YES	YES	26	15		
4.28	35	42	30	34	53	45	60	52	100%	100%	YES	YES	24	15		
4.29	35	43	30	34	53	45	60	52	100%	100%	YES	YES	24	15		

17 Building 302 Block A – Level 5

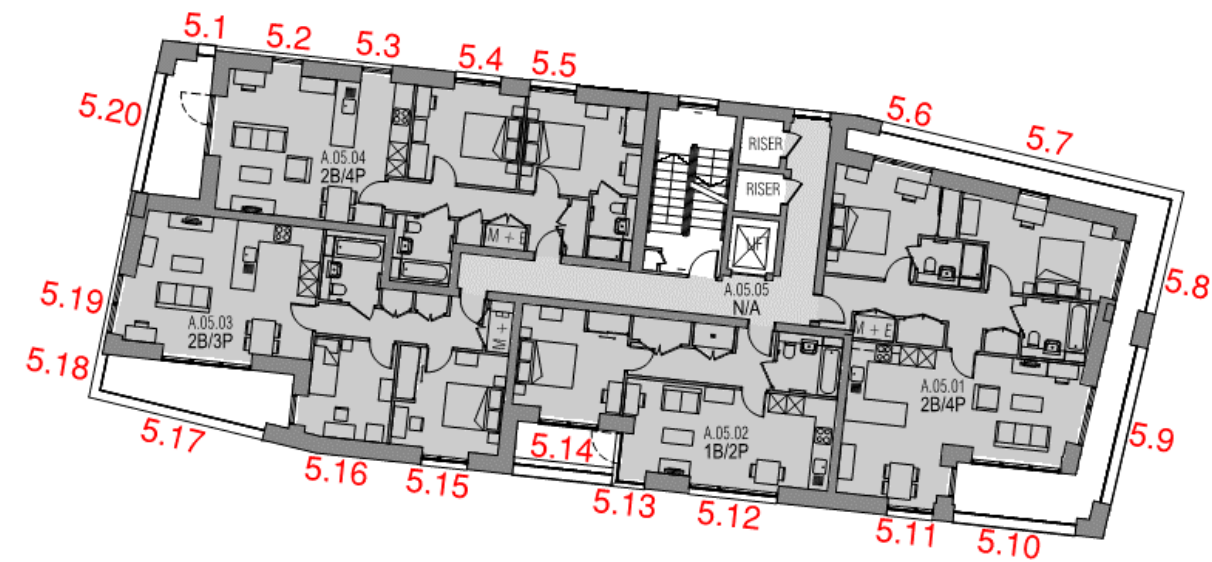


Figure 17 Building 302 Block A – Level 5 floor plan

Table 19 Building 302 Block A – Level 5 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area				Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (L_{Aeq})		Dredgers (L_{A63Hz})		All sources night (L_{Aeq})		All sources day (L_{Aeq})										
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	D_w+C_{tr}	D_{63Hz}			
5.1	46	44	43	36	58	58	60	61	30%	100%	YES	YES	29	18	70% open area with 2 dB low frequency allowance		
5.2	46	44	43	36	58	58	60	61	30%	100%	YES	YES	29	18	Seal and ventilate via balcony or provide Juliet screening arrangement. 70% open area with 2 dB low frequency allowance		
5.3	46	44	43	36	58	58	60	61	30%	100%	YES	YES	29	18	Juliet screening arrangement.		
5.4	46	45	43	41	58	58	60	61	30%	70%	YES	YES	29	18	70% open area with 2 dB low frequency allowance		
5.5	46	45	43	41	58	58	60	61	30%	70%	YES	YES	29	18			
5.6	46	45	42	39	58	58	60	60	45%	85%	YES	YES	29	20	70% open area with 1 dB low frequency allowance		
5.7	46	45	39	39	58	58	60	60	70%	85%	YES	YES	29	20			
5.8	45	45	40	40	55	54	58	56	85%	85%	YES	YES	26	17			
5.9	45	44	41	40	54	52	58	55	70%	85%	YES	YES	25	19			
5.10	43	43	33	33	51	47	58	52	100%	100%	YES	YES	22	15			
5.11	42	43	33	33	51	47	57	52	100%	100%	YES	YES	22	15			
5.12	41	43	33	33	51	46	57	52	100%	100%	YES	YES	22	15			
5.13	41	43	33	33	51	46	57	52	100%	100%	YES	YES	23	15			
5.14	42	43	33	38	51	46	57	52	100%	100%	YES	YES	23	15			
5.15	42	44	33	38	51	47	57	53	100%	100%	YES	YES	23	15			
5.16	41	44	33	38	51	47	57	53	100%	100%	YES	YES	23	15			
5.17	41	44	33	38	51	47	57	53	100%	100%	YES	YES	22	15			
5.18	37	44	33	31	55	55	58	59	85%	100%	YES	YES	26	15			
5.19	37	44	33	31	55	55	58	59	85%	100%	YES	YES	26	15	Juliet screening arrangement		
5.20	42	44	33	31	55	55	58	59	100%	100%	YES	YES	26	15			

18 Building 302 Block D – Level 5

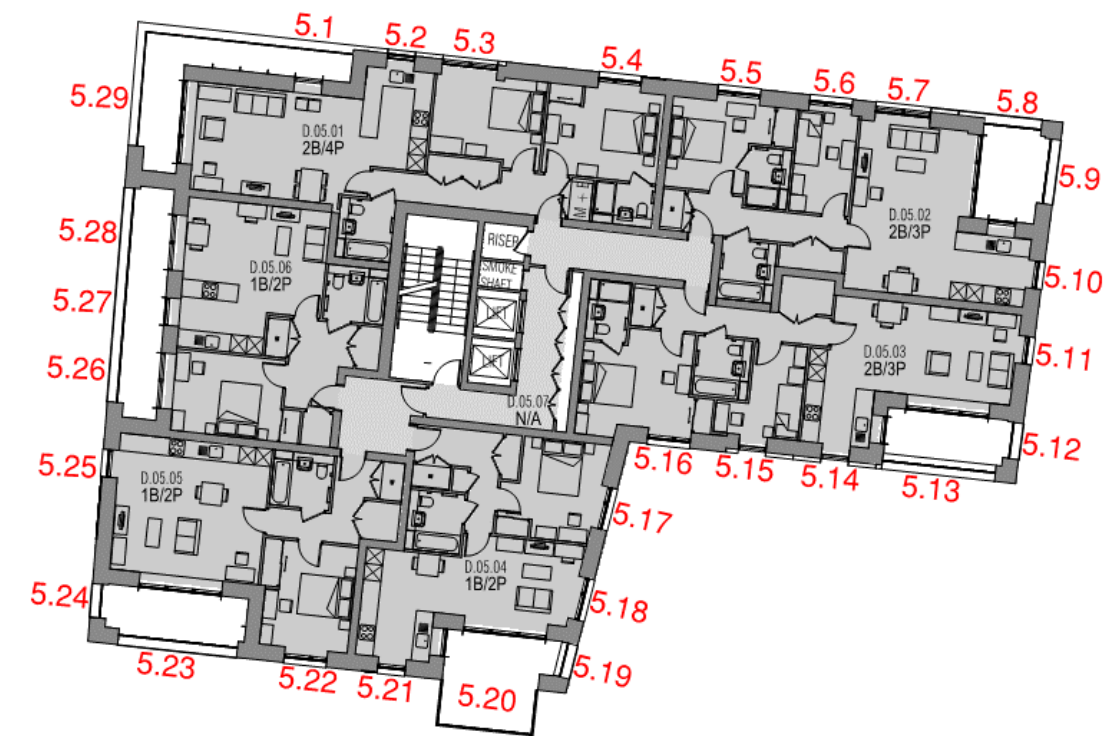


Figure 18 Building 302 Block D – Level 5 floor plan

Table 20 Building 302 Block D – Level 5 predicted noise levels and required mitigation measures

Ref.	Noise levels (dB)								Balcony/ shroud open area				Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (L_{Aeq})		Dredgers (L_{A63Hz})		All sources night (L_{Aeq})		All sources day (L_{Aeq})										
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	D_w+C_{tr}	D_{63Hz}			
5.1	47	48	43	44	52	49	59	53	30%	20%	YES	YES	23	21	Fix window to avoid excess balcony screening. Ventilate via 5.26. Seal and ventilate via balcony or provide Juliet screening arrangement. 30% open area with 1 dB low frequency allowance Juliet screening arrangement. 30% open area with 1 dB low frequency allowance		
5.2	47	48	43	44	52	49	59	53	30%	20%	YES	YES	23	21			
5.3	47	48	43	44	52	49	59	53	30%	20%	YES	YES	25	21			

Ref.	Noise levels (dB)								Balcony/ shroud open area				Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (<i>L</i> _{Aeq})		Dredgers (<i>L</i> _{A63Hz})		All sources night (<i>L</i> _{Aeq})		All sources day (<i>L</i> _{Aeq})										
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	<i>D</i> _w + <i>C</i> _{tr}	<i>D</i> _{63Hz}			
5.4	47	47	43	43	52	49	59	54	30%	30%	YES	YES	25	20	Juliet screening arrangement 45% open area with 1 dB low frequency allowance		
5.5	47	47	43	43	52	49	58	54	30%	30%	YES	YES	24	20			
5.6	47	47	43	43	52	49	58	54	30%	30%	YES	YES	23	20			
5.7	47	47	43	43	52	49	58	54	30%	30%	YES	YES	23	20	Juliet screening arrangement 45% open area with 1 dB low frequency allowance		
5.8	46	47	43	43	52	49	58	54	30%	30%	YES	YES	23	20			
5.9	47	48	43	42	54	50	62	56	30%	30%	YES	YES	25	20			
5.10	47	48	43	43	54	50	62	56	30%	30%	YES	YES	25	20	Seal and ventilate via balcony or provide Juliet screening arrangement		
5.11	48	48	42	42	55	50	65	59	30%	30%	YES	YES	26	19			
5.12	48	48	42	42	55	51	65	59	30%	30%	YES	YES	26	19			
5.13	46	46	33	33	56	49	63	57	70%	70%	YES	YES	27	15	Juliet screening arrangement Juliet screening arrangement Juliet screening arrangement Juliet screening arrangement Seal and ventilate via balcony or provide Juliet screening arrangement.		
5.14	47	44	33	33	56	48	63	57	45%	100%	YES	YES	27	15			
5.15	47	43	33	34	56	49	63	57	45%	100%	YES	YES	27	15			
5.16	47	44	33	34	56	49	63	57	45%	100%	YES	YES	29	15			
5.17	47	44	33	34	56	49	63	57	45%	100%	YES	YES	29	15			
5.18	48	46	35	36	57	50	63	56	30%	70%	YES	YES	29	15	Seal and ventilate via balcony or provide Juliet screening arrangement.		
5.19	48	46	35	36	56	50	63	56	30%	70%	YES	YES	27	15			
5.20	46	47	33	33	56	50	63	57	45%	45%	YES	YES	27	15			
5.21	47	47	33	33	56	50	63	57	45%	45%	YES	YES	27	15	Seal and ventilate via balcony or provide Juliet screening arrangement.		
5.22	46	46	33	33	56	51	63	57	70%	70%	YES	YES	29	15			
5.23	46	46	32	33	56	51	63	57	70%	70%	YES	YES	27	15			
5.24	38	47	33	38	54	50	61	55	100%	45%	YES	YES	25	15	Seal and ventilate via balcony or provide Juliet screening arrangement.		
5.25	38	47	33	38	53	49	60	54	100%	45%	YES	YES	24	15			
5.26	38	46	33	36	53	48	60	54	100%	70%	YES	YES	26	15			
5.27	38	46	33	36	53	48	60	53	100%	70%	YES	YES	24	15			
5.28	38	46	33	36	53	48	60	53	100%	70%	YES	YES	24	15			
5.29	38	44	33	36	53	48	60	53	100%	100%	YES	YES	24	15			